MODULE 11 · DOMAIN 3 - APPLICATION DEVELOPMENT

Choosing an LLM

The attributes that decide the choice, the constraints that disqualify, and why the smallest model that passes usually wins.

SECTION 3 of 6 EXAM WEIGHT 30% MODULE 11 of 40

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Hire for the role, not the CV. The best model is the smallest one that meets every requirement, not the most impressive one available.

Seven attributes that decide the choice

ATTRIBUTE	THE QUESTION TO ASK
Task fit	Chat, code, reasoning, classification, speech, or embeddings?
Inputs	Text only, or images, audio and video too?
Context window	Does everything the request needs fit at once?
Quality	Does it pass evaluation on our own data?
Latency and cost	Fast and cheap enough at our volume?
Terms and data	Licence, data retention and residency acceptable?
Operations	Tool calling, structured output, fine-tuning, throughput?

ON THE EXAM

In a scenario, find the attribute that is stated as a requirement. That one decides the answer. The rest are background.

A bigger model, or a smaller one?

BIGGER MODELS

More capable, more costly

- Better at complex, open tasks
- Slower per token
- Higher cost per request
- Often unnecessary for narrow tasks

SMALLER MODELS

Fast and cheap

- Excellent at well-defined tasks
- Low latency, high throughput
- Much cheaper at volume
- May struggle with complex reasoning

THE TIEBREAKER

Choose the smallest model that passes your evaluation. Paying for capability the task does not use is waste, and the exam treats it that way.

A reasoning model, or not?

REASONING MODELS

Think before answering

- Multi-step problems, planning, maths
- Complex agentic tool use
- More tokens and more latency
- Some always reason and cannot switch it off

NON-REASONING, OR LOW EFFORT

Answer straight away

- Classification, extraction, simple chat
- Fast first token
- Fewer tokens, lower cost
- Enough for most well-defined tasks

THE TIEBREAKER

If the scenario stresses speed on a simple task, a model that always reasons is the wrong choice, however capable it is.

Some jobs need a different kind of model

THE JOB	THE KIND OF MODEL
Write or fix code	A model tuned or benchmarked for code generation
Transcribe audio	A speech recognition model, then an LLM if needed
Understand photos or scans	A model that accepts image inputs
Search by meaning	An embedding model, not a chat model
Follow instructions in a chat	An instruction-tuned model, not a base model

ON THE EXAM

When the input is not text, check the model's supported inputs first. A text-only model cannot see a photo, however large it is.

Open-weight or proprietary?

OPEN-WEIGHT MODELS

Llama, Gemma, Qwen and more

- Weights can be fine-tuned
- Served with provisioned throughput
- Community licence terms apply
- More control over deployment

PROPRIETARY MODELS

Claude, GPT, Gemini and more

- Often the strongest quality
- Provider terms and policies apply
- Some retain data or need cross-geo routing
- Less control, no weight access

THE TIEBREAKER

Fine-tuning your own weights points to open-weight models on provisioned throughput. Frontier quality with no fine-tuning often points to proprietary.

Catalogue details that change the choice

MODEL DETAIL	WHY IT MATTERS
Claude Fable 5, 5.1: 30-day retention	Unusable if you opted out of retention
Some Gemini models: global endpoint	Need cross-geo routing
Some models always reason	Latency you cannot turn off
Claude Sonnet 5 rejects temperature	Setting it returns an error
Gemini 2.5 retires 2 October 2026	Plan migration: 3 months' notice

ON THE EXAM

A constraint that disqualifies beats a benchmark that impresses.

When the window is the deciding attribute

- The window must hold **k chunks, the prompt and the answer**, from module 08.
- When a task needs a **whole long document at once**, a long-context model can beat rag.
- Windows now range from about 128,000 tokens to over a million in the catalogue.
- A bigger window costs more per request and does not fix poor retrieval.
- The LLM's window is not the embedding model's window. Keep them apart.

> ON THE EXAM
>
> Choose a long-context model when the task genuinely needs everything at once, such as cross-referencing a whole contract.

How to choose, in order

01

Disqualify

Drop models failing terms, residency or inputs.

02

Fit

Keep models suited to the task.

03

Evaluate

Score the shortlist on your own data.

04

Pick

The cheapest and fastest that passes.

05

Stay swappable

Model name in config, not hard-coded.

ON THE EXAM

Evaluate before you commit, and keep the model name in configuration. The catalogue will change under you.

Harbourline picks different models for different jobs

01

Claims triage

Batch labels: a small, fast model.

02

Customer chat

Mid-size, fast, in-region.

03

Coverage disputes

Reasoning model for underwriters.

04

Damage photos

A model accepting image inputs.

05

All of them

No cross-geo routing, no retention.

ON THE EXAM

One application, several models. Each job gets the smallest model that meets its requirements.

> An analytics team wants a tool that turns plain English requests into SQL and PySpark, and fixes errors in existing notebooks.
>
> Accuracy of the generated code matters more than cost.

Which kind of model should the engineer prioritise?

A. A small general chat model, to keep latency low

B. A model tuned or benchmarked for code generation

C. An embedding model

D. A speech recognition model

◆ PAUSE AND DECIDE

Which kind of model should the engineer prioritise?

A Small general chat model

Optimises latency, which was not the stated priority.

B Code-generation model

Task fit is code, and accuracy is the priority.

C Embedding model

Produces vectors, not code.

D Speech recognition

The input is text, not audio.

THE SIGNAL

Accuracy of the generated code matters more than cost. Task fit plus the stated priority settle it.

> A contact centre wants a summary of every recorded customer call, stored alongside the call record.
>
> The recordings are audio files. Nobody currently transcribes them.

Which approach fits?

A. A text-only LLM with a very long context window for the calls

B. An embedding model run over the audio files to index them

C. An image understanding model applied to each call's waveform

D. Speech recognition to transcribe, then an LLM to summarise

◆ PAUSE AND DECIDE

Which approach fits?

A Long-context text model

Cannot hear audio. Window size is irrelevant.

B Embeddings over audio

Vectors are not summaries, and text embedders take text.

C Image model

Wrong input type.

D Transcribe, then summarise

Speech to text, then a text task. Two steps, two models.

THE SIGNAL

The recordings are audio, and nobody transcribes them. Check the input type before anything else.

> A retailer labels 5 million short product reviews into five categories every night. The budget is tight.
>
> On the team's evaluation set, a small model scores 96% accuracy and a frontier reasoning model scores 97%. The business needs 95%.

Which model should the team choose?

A. The small model, since it passes the evaluation threshold

B. The frontier reasoning model, since it scores highest

C. The frontier model at maximum reasoning effort

D. Fine-tune the largest available open model first

◆ PAUSE AND DECIDE

Which model should the team choose?

A Small model

Passes 95%, fast, and far cheaper across 5 million rows.

B Frontier reasoning model

One point better, at many times the cost. Wasteful.

C Maximum reasoning effort

Even more tokens for a simple labelling task.

D Fine-tune the largest model

Cost and effort to beat a bar already met.

THE SIGNAL

The business needs 95%. Once a model passes, compare cost and speed, not the last point of accuracy.

> A bank's policy requires that no model provider retains prompts or responses. The bank has opted out of data retention.
>
> The team's preferred model retains prompts for 30 days for safety review.

What should the engineer do?

A. Use the preferred model, adding a system prompt about confidentiality

B. Use the preferred model after removing customer names from prompts

C. Choose a model whose data terms meet the no-retention policy

D. Use the preferred model only for internal staff

◆ PAUSE AND DECIDE

What should the engineer do?

A Confidentiality prompt

A prompt cannot change a provider's retention terms.

B Remove names first

Still retained. The policy says no retention, not less.

C Compliant model

The terms disqualify the preferred model. Choose another.

D Internal staff only

The policy has no internal exception.

THE SIGNAL

No provider retains prompts. A hard constraint removes a model from the shortlist. No workaround brings it back.

> Lawyers need to ask questions that cross-reference clauses across entire 300-page contracts, about 200,000 tokens each.
>
> A rag prototype keeps missing answers that depend on clauses far apart in the document.

Which model attribute matters most here?

A. A 128,000-token model, used with larger retrieval chunks

B. A smaller, faster model so that answers come back sooner

C. A context window large enough to hold the whole contract

D. An embedding model with a longer window for the clauses

◆ PAUSE AND DECIDE

Which model attribute matters most here?

A 128k model, bigger chunks

200,000 tokens do not fit in 128,000. Still split.

B Smaller model

Speed was not the problem.

C Window holds the whole contract

No splitting, so distant clauses are seen together.

D Longer embedding window

Improves retrieval, but still retrieves pieces.

THE SIGNAL

Cross-reference clauses across entire contracts. When the task needs everything at once, the context window decides.

> A help widget answers simple FAQ questions, and users expect the first words to appear in under a second.
>
> The team chose a model that always reasons before answering. Answers are correct, but users complain about the wait.

What should the engineer change?

A Switch to a fast non-reasoning model, or low reasoning effort

B Increase the maximum output tokens so replies can finish

C Add a reranker to retrieval so the best FAQ comes first

D Move to a larger reasoning model that thinks more quickly

◆ PAUSE AND DECIDE

What should the engineer change?

A Fast or low-effort model

Simple task, speed required: remove the reasoning delay.

B More output tokens

Allows longer answers. Does not start them sooner.

C Add a reranker

Adds latency, the opposite of what is needed.

D Larger reasoning model

Slower still, for a task that needs no reasoning.

THE SIGNAL

Simple FAQ questions, under a second, and a model that always reasons. The reasoning is the delay.

> An insurer wants an assistant that reads a claimant's written description together with photos of the damage.
>
> It must describe what the photos show and flag any mismatch with the written account.

Which model attribute is essential?

A. A large text-only model with a long context window

B. An embedding model with high dimensions

C. OCR on the photos, then a text-only model

D. Support for image inputs alongside text

◆ PAUSE AND DECIDE

Which model attribute is essential?

A Large text-only model

Cannot see the photos at all.

B High-dimension embeddings

Vectors, not descriptions or comparisons.

C OCR then text model

OCR reads printed words. A dented bumper has none.

D Image inputs

The model must see the damage to describe it.

THE SIGNAL

Describe what the photos show. The input is an image, so the model must accept images.

> An EU insurer's data residency policy forbids processing outside its region, so cross-geography processing is disabled in its workspace.
>
> The model the team wants is hosted on a global endpoint that requires cross-geography routing.

What should the engineer do?

A Enable cross-geography routing for this one application

B Choose a model available in-region without cross-geography routing

C Call the same model through ai_query instead

D Call the same model through a different client library

◆ PAUSE AND DECIDE

What should the engineer do?

A Enable cross-geo routing

Breaks the residency policy the scenario states.

B In-region model

Meets the residency constraint. Disqualify, then choose.

C Through ai_query

Same model, same global endpoint, same routing.

D Different client library

Changes the code, not where data is processed.

THE SIGNAL

Forbids processing outside its region. Residency is a hard constraint: it removes the model, whatever the access path.

What to carry forward

- Seven attributes: **task, inputs, context, quality, latency and cost, terms and data, operations**.
- Disqualify first on hard constraints: retention, residency, inputs, licence.
- Then choose the **smallest model that passes** your own evaluation.
- Reasoning models for multi-step problems. Not for fast, simple answers.
- Long context when a task needs a whole document at once. Keep the model name in config.

> ON THE EXAM
>
> Next: module 12 reads model cards and hubs, and picks a model from experiment metrics.